

HW #3

Question 1: Image restoration (60%)

1.1 (30%)

(a) Write a MATLAB code for generating a degraded image $g(x, y)$ from the input $f(x, y)$, where the degradation is caused by a linear motion blur and additive Gaussian noise as $g(x, y) = h(x, y) \star f(x, y) + \eta(x, y)$.

It is known that the linear motion blur comes from a horizontal translation of the image by 7 pixels and a rotation by 45 degree. Also, the Gaussian noise has zero mean and variance of 0.01, for pixel values defined in the interval $[0, 1]$. As input use the image provided on the course website ([building.pgm](#)).

(b) Write a MATLAB code for reconstructing $f(x, y)$ as $\hat{f}(x, y) = h_W(x, y) \star g(x, y)$ by using the Wiener filter, $h_W(x, y)$, where the parameters of the image degradation are known and equal to those given in (a).

Submit the listing of your MATLAB code, Figure 1 showing the degraded image from (a), and Figure 2 showing the reconstructed image from (b).

Hint:

Use the following commands:

```
'fspecial('motion',...)',  
'imfilter()',  
'imnoise('gaussian',...)',  
'deconvwnr()'
```

1.2 (30%)

[urbanblur.png](#) is a blurred version [urban0.png](#). Use Wiener filter to restore the blurred image.

Write out the transfer function of the filter being used. Guess and try a few filter parameters, present the resulting pictures, compute their mean-square-error from the original image, and discuss what you see.



urbanblur.png



urban0.png

Question 2: Color Segmentation (40%)

Please write programs to automatically segment and display the following image into 5 images with all the words in color of red, blue, orange, purple, and green respectively. For example, image 1 should contain all the words in red color.

red blue orange purple

orange blue green red

blue purple green red

orange blue red green

purple orange red blue

green red blue purple

orange blue red green

green purple orange red